

Node	Type of Node	Target Metric	Type of Issue	Relevance	Traffic Impacted	Parameter Range	Total Count	Description
petInfo	database	availability	traffic spike / throttling	as practitioners we have run into this type of issue	a fraction of traffic will be throttled	-	6	We generate many read requests directly hitting the database, triggering throttling.
petSearch	containerized micro-service on ECS	availability	human configuration error	configurations are often done manually and cannot be tested as easily as code	random fraction of all traffic	1%, 2%, 3%	10	A random fraction of requests attempts to access a directory in the file system that does not exist.
petSearch	containerized micro-service on ECS	latency	performance degradation	performance degradations can happen due to deploying less performant code, change in hardware, overall resource exhaustion	bunny requests	0.5sec, 1sec, 2sec	10	Bunny requests are delayed by some number of seconds.
payforadoption	containerized micro-service on ECS	availability	memory leak	among the most common issues in cloud systems, see Mariani et al. (2018) ; Ikram et al. (2022)	bunny requests	n/a	6	A memory leak is triggered for all requests.
payforadoption	containerized micro-service on ECS	latency	performance degradation	performance degradations can happen due to deploying less performant code, change in hardware, overall resource exhaustion	all	0.25sec, 0.5sec, 1sec	10	All requests are delayed by some number of seconds.
statusupdater	serverless data-processing system on lambda	availability	human configuration error	configurations are often done manually and cannot be tested as easily as code	kitten requests	n/a	6	Kitten requests attempt to access a database table that does not exist.
statusupdater	serverless data-processing system on lambda	latency	performance degradation	performance degradations can happen due to deploying less performant code, change in hardware, overall resource exhaustion	all requests	0.25sec, 0.5sec, 1sec	10	All requests are delayed by some number of seconds.

Table 4: Breakdown of injected issues. In all cases, issues were repeated twice.